

to the ARM instruction set on 32-bit memory. In ARMv7, this goal can be said to have been met.

**Jazelle.** Jazelle direct byte-code execution (DBX) is a technique that allows Java byte-code to be executed directly in the ARM architecture as a third execution state, alongside existing ARM and Thumb modes.

**NEON.** The advanced SIMD extension (NEON) is a combined 64-bit and 128-bit SIMD instruction set that provides standardised acceleration for media and signal-processing applications. NEON is included in all Cortex-A8 devices but is optional in Cortex-A9 devices. It features a comprehensive instruction set, separate register files and independent execution hardware.

**TrustZone.** This provides a low-cost alternative to adding an additional dedicated security core to an SoC by providing two virtual processors backed by hardware based access control. This lets the

application core switch between two states, referred to as worlds, in order to prevent information from leaking from the more-trusted world to the less-trusted world.

## ARMv8

ARMv8-A represents a fundamental change to ARM architecture. It adds a 64-bit architecture called AArch64 and a new A64 instruction set. AArch64 provides user-space compatibility with AArch32 and A32. Thumb instruction sets have no 64-bit counterpart. It allows 32-bit applications to be executed in a 64-bit operating system (OS), and a 32-bit OS to be under the control of a 64-bit hypervisor.

ARMv8 architecture has the following features:

1. New instruction set: A64
2. Thirty one general-purpose 64-bit registers
3. Dedicated stack pointer
4. Program counter is no longer

accessible as a register

5. Most instructions can take 32-bit or 64-bit arguments

6. Addresses assumed to be 64-bit

7. Advanced SIMD (NEON) is enhanced

8. New exception system

## Advantages of ARMv8 over ARMv7

ARMv8 uses program-counter relative addressing. The program counter and stack pointer are no more general-purpose registers, thereby increasing the number of general-purpose registers from 14 to 31. More registers reduce the need for register-to-stack copies. Double-precision vectors with IEEE support have been added. The peculiar interrupt modes and banked registers are mostly gone. The load/store multiple instructions have been replaced with load/store pairs. NEON has also improved. For the smartphone world, ARMv8 offers support of more than 4GB RAM. ●

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